Why a Jury of Judges Cuts Bias $\sim 1/\sqrt{n}$, Worked Out by Hand

Averaging judges shrinks noise like $1/\sqrt{n}$, majority vote beats one judge — and correlated bias breaks both.

What this document does. It shows two complementary facts about using several LLM judges instead of one. First, averaging n independent judge scores divides the noise variance by n, so the error falls as $1/\sqrt{n}$. Second, a majority vote of n judges that are each right with probability $q > 0.5$ is more reliable than one (Condorcet). Then it delivers the warning: if the judges share a bias, they are not independent, and a variance floor stops the averaging from helping. Every number is reproduced by the program in Appendix A.

Where this sits. This is **Topic 2 of Module 3.6** (LLM-as-judge, bias mitigation). Companions: Module 3.6 Topic 1 (judge validation, position bias) and Module 3.5 (sample size).

Concepts covered, in order: variance of an average · the $1/\sqrt{n}$ law · the binomial majority (Condorcet) · diminishing returns vs linear cost · the correlated-bias variance floor.

Internal learning material. Numbers are reproducible from Appendix A. v1.0

1 The setup: many judges, one verdict

A single LLM judge is noisy and biased. A *jury* runs n judges and combines them — by averaging their scores, or by majority vote on a binary verdict. We quantify exactly how much that helps under two regimes: independent judges (the ideal) and correlated judges (the reality).

Quantity	Symbol	Value here
Jury size	n	$1, 3, 5, 7, \ldots$
Single-judge score noise (std)	σ	1.0
Single-judge correctness prob	q	0.70
Inter-judge error correlation	ρ	0 (ideal) / 0.30 (real)

Notation. Each judge i reports score $X_i = \mu + \varepsilon_i$ with $\mathrm{Var}(\varepsilon_i) = \sigma^2$. The jury score is the mean $\bar{X} = \frac{1}{n}\sum_i X_i$. For votes, each judge is correct independently with probability q and we take the majority.

Intuition

One judge's error is a single draw from a noisy, possibly skewed distribution. Averaging several *independent* judges lets their random errors partly cancel — some high, some low — so the mean lands closer to the truth. The catch, foreshadowed now: errors only cancel if they are independent. Shared bias points the same way every time and never cancels.

2 Step 1 — the variance of an average

For n independent judges each with variance σ^2,

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}, \qquad \mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}. \tag{1}$$

So the standard error falls as $1/\sqrt{n}$. With $\sigma = 1$:

n	$\mathrm{SE} = \sigma/\sqrt{n}$	noise vs 1 judge	reading
1	1.000	1.00×	baseline
4	0.500	0.50×	4 judges halve the noise
9	0.333	0.33×	9 judges third it

✓ To halve the error you need 4× the judges; to third it, 9×. The benefit is real but sub-linear while the cost is strictly linear in n.

> **Mini-example — this operation in isolation**
> Why n^2 in the denominator becomes n: variance of a *sum* of n independent terms is $n\sigma^2$; dividing the sum by n to make a mean divides its variance by n^2. Net: $n\sigma^2/n^2 = \sigma^2/n$. The square root turns the $1/n$ on variance into $1/\sqrt{n}$ on the error you actually feel.

3 Step 2 — the majority vote (Condorcet)

For a binary verdict, combine by majority. If each of n (odd) judges is correct independently with probability q, the jury is correct when at least $\lceil n/2 \rceil$ agree:

$$P_{\mathrm{maj}}(n,q) = \sum_{k=\lfloor n/2\rfloor+1}^{n}\binom{n}{k}q^k(1-q)^{n-k}. \tag{2}$$

With $q = 0.70$:

n	P_{maj}	gain over a single judge
1	0.7000	—
3	0.7840	+0.084
5	0.8369	+0.053
7	0.8740	+0.037

✓ (Check $n = 3$: $q^3 + 3q^2(1-q) = 0.343 + 3(0.49)(0.30) = 0.343 + 0.441 = 0.784$.) Each pair of extra judges helps less than the last — diminishing returns again. This is Condorcet's jury theorem: for $q > 0.5$, more voters monotonically improve the majority, approaching certainty as $n \to \infty$.

$$\boxed{q > 0.5 \Rightarrow P_{\mathrm{maj}}(n,q) \text{ rises with } n; \quad q < 0.5 \Rightarrow \text{it falls.}} \tag{3}$$

Watch out

Condorcet requires $q > 0.5$. If your judges are worse than a coin flip on some axis (e.g. a shared length bias makes them *systematically* prefer the wrong answer), majority voting makes it *worse* — it amplifies the wrong consensus. Validate $q > 0.5$ per axis before trusting a jury.

4 Step 3 — correlated bias puts a floor under the noise

The $1/\sqrt{n}$ law assumed independence. If judge errors share a common bias with correlation ρ, the variance of the mean becomes

$$\mathrm{Var}(\bar{X}) = \sigma^2\left(\rho + \frac{1-\rho}{n}\right) \xrightarrow[n\to\infty]{} \rho\,\sigma^2. \tag{4}$$

The independent part still averages away as $1/n$, but the shared part $\rho\sigma^2$ never does. With $\rho = 0.30, \sigma = 1$:

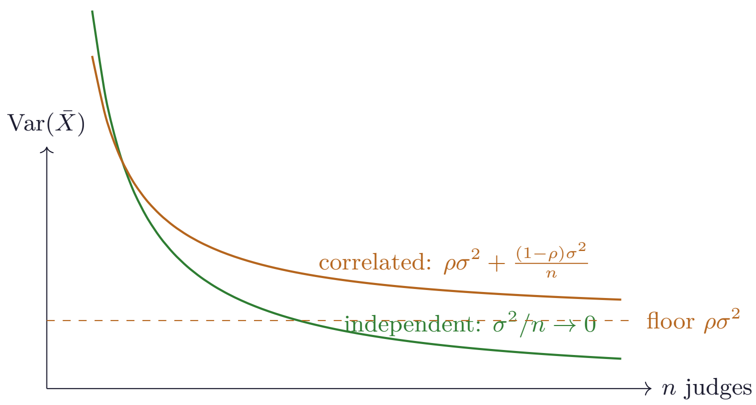
n	$\mathrm{Var}(\bar{X})$ at ρ=0.3	vs ideal σ^2/n
3	0.5333	ideal 0.333
5	0.4400	ideal 0.200
100	0.3070	ideal 0.010

At $n = 100$ the ideal variance would be 0.010, but correlation pins it at 0.307 — a floor of $\rho\sigma^2 = 0.30$. ✓ You can add judges forever and never beat it.

Intuition

Correlated judges are really one judge wearing n masks. Diversity — different models, different prompts, swapped positions, varied rubrics — is what lowers ρ and unlocks the $1/\sqrt{n}$ gain. A jury of five copies of the same model with the same prompt buys far less than the formula promises. Spend on *independence*, not just on count.

5 The two regimes, drawn



Independent error chases zero; correlated error flattens onto the $\rho\sigma^2$ floor.

6 Cost cheat-sheet

Variance of mean (indep.)	σ^2/n
Standard error	$\sigma/\sqrt{n}$ (halve noise $\Rightarrow 4\times$ judges)
Majority correct	$\sum_{k>n/2} \binom{n}{k} q^k (1-q)^{n-k}$
Condorcet	$q > 0.5 \Rightarrow$ more judges better
Correlated variance	$\sigma^2(\rho + \frac{1-\rho}{n})$
Noise floor	$\rho\sigma^2$ (never beaten)

7 An honest word about these numbers

$\sigma = 1$, $q = 0.70$, $\rho = 0.30$ are illustrative. What is invariant: independent judge noise falls as $1/\sqrt{n}$, a majority of better-than-chance judges beats one (Condorcet), and shared bias imposes a hard variance floor $\rho\sigma^2$ that no jury size removes. The practical lesson is the floor: the return on a jury comes from making the judges *different*, not merely numerous.

Appendix A — Reference implementation

```python
# jury_voting.py -- reproduces every number in "Why a Jury of Judges...".
import math
from math import comb

sigma = 1.0
# 1/sqrt(n) standard error
for n in (1, 4, 9):
    print(f"n={n} SE={sigma/math.sqrt(n):.4f}")

# Condorcet majority correctness at q=0.7
def majority(n, q):
    return sum(comb(n, k) * q**k * (1-q)**(n-k) for k in range(n//2 + 1, n + 1))
for n in (1, 3, 5, 7):
    print(f"n={n} P_majority(q=0.7)={majority(n, 0.7):.4f}")

# correlated variance of the mean
rho = 0.30
for n in (3, 5, 100):
    print(f"n={n} Var_corr={sigma**2*(rho + (1-rho)/n):.4f} (floor={rho*sigma**2})")

# Expected output:
# n=1 SE=1.0000
# n=4 SE=0.5000
# n=9 SE=0.3333
# n=1 P_majority(q=0.7)=0.7000
# n=3 P_majority(q=0.7)=0.7840
# n=5 P_majority(q=0.7)=0.8369
# n=7 P_majority(q=0.7)=0.8740
# n=3 Var_corr=0.5333 (floor=0.3)
# n=5 Var_corr=0.4400 (floor=0.3)
# n=100 Var_corr=0.3070 (floor=0.3)
```

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